

Layer-Wise Selection of B-Spline Degree and Grid Size in Kolmogorov–Arnold Networks

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1 Setting and Notation

Consider a Kolmogorov–Arnold network (KAN) [1] of shape $[n_0, n_1, \dots, n_L]$. Layer $l \in \{0, \dots, L-1\}$ maps $x_l \in \mathbb{R}^{n_l}$ to $x_{l+1} \in \mathbb{R}^{n_{l+1}}$ by

$$x_{l+1,j} = \sum_{i=1}^{n_l} \phi_{l,j,i}(x_{l,i}), \quad j = 1, \dots, n_{l+1}. \quad (1)$$

In the trained network each edge function is a B-spline of degree k_l on a uniform grid of G_l intervals covering $[a_l, b_l]$,

$$\hat{\phi}_{l,j,i}(x) = \sum_{n=0}^{G_l+k_l-1} c_{l,j,i,n} B_n^{k_l}(x), \quad h_l = \frac{R_l}{G_l}, \quad R_l = b_l - a_l, \quad (2)$$

so that $\hat{\phi}_{l,j,i} \in C^{k_l-1}$. Here k_l denotes the polynomial degree ($k_l = 3$ is the cubic spline used in [1]). Layer l has $w_l = n_l n_{l+1}$ edges, and the number of trainable spline coefficients is

$$P = \sum_{l=0}^{L-1} w_l (G_l + k_l). \quad (3)$$

Assumption 1. The target f admits an exact representation of the form (1) with edge functions $\phi_{l,j,i}$, and all layer inputs of both f and the approximating network $\hat{f}$ lie in $[a_l, b_l]$. The second condition holds when inputs are normalized between layers (e.g., by LayerNorm or tanh) or when the grid is extended to cover the input range.

For each layer define

$$\Lambda_l = \max_{j,i} \|\phi'_{l,j,i}\|_\infty, \quad \Lambda'_l = \max_{j,i} \|\phi''_{l,j,i}\|_\infty, \quad M_l^{(q)} = \max_{j,i} \|\phi_{l,j,i}^{(q)}\|_\infty, \quad (4)$$

and the per-layer approximation errors

$$\delta_l = \max_{j,i} \|\phi_{l,j,i} - \hat{\phi}_{l,j,i}\|_\infty, \quad \delta'_l = \max_{j,i} \|\phi'_{l,j,i} - \hat{\phi}'_{l,j,i}\|_\infty. \quad (5)$$

All norms are taken over $[a_l, b_l]$. For vectors $\|\cdot\|_\infty$ is the maximum norm and for matrices the induced maximum row-sum norm.

Spline approximation. For $g \in C^q[a, b]$ with $1 \leq q \leq k + 1$, there is a spline s of degree k on a uniform grid of spacing h (a quasi-interpolant) that satisfies simultaneously [2, 3]

$$\|g - s\|_\infty \leq C_{k,q} h^q \left\| g^{(q)} \right\|_\infty, \quad (6)$$

$$\|g' - s'\|_\infty \leq C'_{k,q} h^{q-1} \left\| g^{(q)} \right\|_\infty, \quad (7)$$

with constants independent of h and g . The best attainable exponents are therefore $k + 1$ for the values and k for the derivative, and they are reduced to q and $q - 1$ when g has only q bounded derivatives.

2 Propagation of Value Error

Theorem 1. Under Assumption 1, let $e_l = \|x_l - \hat{x}_l\|_\infty$ with $e_0 = 0$. Then

$$e_{l+1} \leq n_l \Lambda_l e_l + n_l \delta_l, \quad (8)$$

and consequently

$$\|f - \hat{f}\|_\infty \leq \sum_{l=0}^{L-1} A_l \delta_l, \quad A_l = n_l \prod_{m=l+1}^{L-1} n_m \Lambda_m. \quad (9)$$

Proof. For output unit j of layer l ,

$$x_{l+1,j} - \hat{x}_{l+1,j} = \sum_{i=1}^{n_l} [\phi_{l,j,i}(x_{l,i}) - \phi_{l,j,i}(\hat{x}_{l,i})] + \sum_{i=1}^{n_l} [\phi_{l,j,i}(\hat{x}_{l,i}) - \hat{\phi}_{l,j,i}(\hat{x}_{l,i})]. \quad (10)$$

By the mean value theorem each term of the first sum is at most $\Lambda_l e_l$ in magnitude, and by Assumption 1 each term of the second sum is at most δ_l . Taking the maximum over j gives (8). Unrolling (8) from $e_0 = 0$ gives $e_L \leq \sum_l n_l \delta_l \prod_{m=l+1}^{L-1} n_m \Lambda_m$, which is (9). $\square$

The factor A_l is the amplification of an error made in layer l by all layers after it; $A_{L-1} = n_{L-1}$. Combining (9) with (6) for $q = k_l + 1$ gives

$$\|f - \hat{f}\|_\infty \leq \sum_{l=0}^{L-1} A_l C_{k_l} M_l^{(k_l+1)} \left(\frac{R_l}{G_l} \right)^{k_l+1}, \quad (11)$$

which for equal grids recovers the rate $G^{-(k+1)}$ of [1, Thm. 2.1]. In (11) each layer contributes at the rate of its own degree.

3 Propagation of Jacobian Error

Write $D_l(x_l) \in \mathbb{R}^{n_{l+1} \times n_l}$ for the Jacobian of layer l , with entries $(D_l)_{ji} = \phi'_{l,j,i}(x_{l,i})$, and $\hat{D}_l$ for the same quantity of $\hat{f}$ evaluated at $\hat{x}_l$. By the chain rule,

$$J_f = D_{L-1} D_{L-2} \cdots D_0, \quad J_{\hat{f}} = \hat{D}_{L-1} \hat{D}_{L-2} \cdots \hat{D}_0. \quad (12)$$

Theorem 2. Under Assumption 1 with $\phi_{l,j,i} \in C^2$, let

$$\eta_l = n_l (\Lambda'_l e_l + \delta'_l). \quad (13)$$

Then $\|D_l - \hat{D}_l\|_\infty \leq \eta_l$, and

$$\|J_f - J_{\hat{f}}\|_\infty \leq \sum_{l=0}^{L-1} \left(\prod_{m=l+1}^{L-1} n_m \Lambda_m \right) \eta_l \prod_{m=0}^{l-1} (n_m \Lambda_m + \eta_m). \quad (14)$$

Proof. For each entry,

$$|\phi'_{l,j,i}(x_{l,i}) - \hat{\phi}'_{l,j,i}(\hat{x}_{l,i})| \leq |\phi'_{l,j,i}(x_{l,i}) - \phi'_{l,j,i}(\hat{x}_{l,i})| + |\phi'_{l,j,i}(\hat{x}_{l,i}) - \hat{\phi}'_{l,j,i}(\hat{x}_{l,i})| \leq \Lambda'_l e_l + \delta'_l. \quad (15)$$

Each row has n_l entries, so $\|D_l - \hat{D}_l\|_\infty \leq \eta_l$. In addition $\|D_m\|_\infty \leq n_m \Lambda_m$ and $\|\hat{D}_m\|_\infty \leq n_m \Lambda_m + \eta_m$. The telescoping identity

$$D_{L-1} \cdots D_0 - \hat{D}_{L-1} \cdots \hat{D}_0 = \sum_{l=0}^{L-1} D_{L-1} \cdots D_{l+1} (D_l - \hat{D}_l) \hat{D}_{l-1} \cdots \hat{D}_0 \quad (16)$$

and submultiplicativity of the induced norm give (14). $\square$

Corollary 1 (weakest-link degree). *Let all grids share the spacing h and let each $\phi_{l,j,i} \in C^{k_l+1}$. Then*

$$\|J_f - J_{\hat{f}}\|_\infty = O(h^\kappa), \quad \kappa = \min_{0 \leq l \leq L-1} k_l. \quad (17)$$

Proof. By (6)–(7), $\delta_l = O(h^{k_l+1})$ and $\delta'_l = O(h^{k_l})$. By (8), $e_l = O(h^{\min_{m < l} (k_m+1)})$, and $\min_{m < l} (k_m + 1) > \kappa$. Hence $\eta_l = O(h^{k_l}) + O(h^{\kappa+1})$, and the slowest term in (14) is $O(h^\kappa)$. $\square$

The exponent in (17) is attained for generic edge functions, since for $L = 1$ the Jacobian error equals δ'_0 , which is of exact order h^{k_0} whenever $\phi^{(k_0+1)} \neq 0$. The asymmetry with (11) is the main structural point: value accuracy is limited layer by layer, whereas derivative accuracy is limited by the lowest degree anywhere in the network. A single piecewise-linear layer ($k_l = 1$) restricts the Jacobian error to $O(h)$ and makes $J_{\hat{f}}$ discontinuous, regardless of the degrees used elsewhere.

4 Degree Selection per Layer

Proposition 1 (smoothness saturation). *If the edge functions of layer l have r_l bounded derivatives and no more (in the Sobolev sense, $\phi_{l,j,i} \in W^{r_l, \infty}$), then*

$$\delta_l = O(h_l^{\min(k_l+1, r_l)}), \quad \delta'_l = O(h_l^{\min(k_l, r_l-1)}). \quad (18)$$

Proof. Apply (6)–(7) with $q = \min(k_l + 1, r_l)$; for non-integer r_l the same bounds hold with $q = r_l$ by the corresponding Besov–Sobolev estimates for splines [2, 3]. $\square$

Raising k_l above $\lceil r_l \rceil - 1$ therefore gives no gain in rate, while it adds w_l coefficients per unit of degree by (3) and widens each basis function to $k_l + 1$ intervals. If the task requires the derivatives of $\hat{f}$ up to order v , Corollary 1 adds a lower bound. Together,

$$v \leq k_l \leq \max(v, \lceil r_l \rceil - 1), \quad l = 0, \dots, L-1. \quad (19)$$

For localization, the estimator $\hat{\theta} = \hat{f}(I)$ of emitter position from a Poisson image I with mean $\mu(\theta)$ has, to first order,

$$\text{Cov}(\hat{\theta}) \approx J_{\hat{f}} \text{diag}(\mu(\theta)) J_{\hat{f}}^\top, \quad (20)$$

so the variance, and with it the gap to the Cramér–Rao lower bound [5], is governed by the Jacobian. Localization networks therefore have $v = 1$ and need $k_l \geq 1$ in every layer, with $k_l \geq 2$ for a continuous Jacobian. The pixel-integrated Gaussian PSF is analytic, so r_l is large and the upper bound in (19) does not bind; cubic splines in every layer are the natural choice.

5 Optimal Grid Allocation

Fix the degrees $k_l = k$ and write $p = k + 1$. Ignoring the constant offset k in (3), a coefficient budget B imposes $\sum_l w_l G_l = B$. The bound (11) becomes

$$E(G_0, \dots, G_{L-1}) = \sum_{l=0}^{L-1} \alpha_l G_l^{-p}, \quad \alpha_l = A_l C_k M_l^{(p)} R_l^p. \quad (21)$$

Theorem 3. *The minimizer of (21) over $G_l > 0$ subject to $\sum_l w_l G_l = B$ is*

$$G_l^* = \frac{B}{S} \left(\frac{\alpha_l}{w_l} \right)^{1/(p+1)}, \quad S = \sum_{m=0}^{L-1} \alpha_m^{1/(p+1)} w_m^{p/(p+1)}, \quad (22)$$

with minimum value

$$E^* = \frac{S^{p+1}}{B^p}. \quad (23)$$

The uniform grid $G_l = B/W$, $W = \sum_l w_l$, gives $E_u = W^p \sum_l \alpha_l / B^p$, and

$$\frac{E^*}{E_u} = \frac{S^{p+1}}{W^p \sum_l \alpha_l} \leq 1, \quad (24)$$

with equality if and only if α_l / w_l is the same for every layer.

Proof. Each term $\alpha_l G_l^{-p}$ is strictly convex on $G_l > 0$ and the constraint is linear, so the Karush–Kuhn–Tucker conditions are necessary and sufficient [4]. Stationarity of the Lagrangian requires

$$p \alpha_l G_l^{-(p+1)} = \lambda w_l, \quad l = 0, \dots, L-1, \quad (25)$$

so $G_l = c (\alpha_l / w_l)^{1/(p+1)}$ for a common $c > 0$. The constraint gives $cS = B$, which is (22). Substituting,

$$E^* = \sum_l \alpha_l c^{-p} \left(\frac{\alpha_l}{w_l} \right)^{-p/(p+1)} = c^{-p} \sum_l \alpha_l^{1/(p+1)} w_l^{p/(p+1)} = c^{-p} S = \frac{S^{p+1}}{B^p}. \quad (26)$$

For (24), Hölder's inequality with exponents $p+1$ and $(p+1)/p$ gives

$$S = \sum_l \alpha_l^{1/(p+1)} w_l^{p/(p+1)} \leq \left(\sum_l \alpha_l \right)^{1/(p+1)} \left(\sum_l w_l \right)^{p/(p+1)}, \quad (27)$$

with equality if and only if $\alpha_l \propto w_l$. Raising both sides to the power $p+1$ completes the proof. $\square$

Two consequences are useful in practice. First, (25) implies $\alpha_l (G_l^*)^{-p} = (\lambda/p) w_l G_l^*$: at the optimum each layer's share of the error bound is proportional to its share of the coefficient budget. Second, the ratio of grids between two layers,

$$\frac{G_l^*}{G_m^*} = \left(\frac{A_l M_l^{(p)} R_l^p w_m}{A_m M_m^{(p)} R_m^p w_l} \right)^{1/(p+1)}, \quad (28)$$

does not depend on the unknown constant C_k . A layer should receive a finer grid when the layers after it amplify errors strongly (large A_l), when its edge functions are rough (large $M_l^{(p)}$), or when it has few edges (small w_l). The quantities A_l and $M_l^{(p)}$ can be estimated from the edge functions of a pilot model by finite differences, for example after grid extension [1].

6 Numerical Verification

The results above are bounds on approximation error, so they are tested here without gradient training: each known edge function is projected onto the spline space (2) by least squares on uniform knots, and errors are measured on 2×10^4 equispaced points (one dimension) or 10^5 uniformly sampled points (four dimensions). Observed rates are least-squares slopes of $\log(\text{error})$ against $\log G$ over the four finest grids. The code and raw results accompany this note.

Convergence rates. Two one-dimensional targets are used: the pixel-integrated Gaussian PSF profile

$$g(x) = \frac{1}{2} \left[\operatorname{erf}\left(\frac{x+1/2}{\sqrt{2}\sigma}\right) - \operatorname{erf}\left(\frac{x-1/2}{\sqrt{2}\sigma}\right) \right], \quad \sigma = 0.6, x \in [-3, 3], \quad (29)$$

which is analytic, and $r(x) = |x|^{5/2}$ on $[-1, 1]$, which has $r = 5/2$ in (18). Table 1 and Figure 1 confirm the exponents $k+1$ and k for the analytic profile and the saturation at $5/2$ and $3/2$ for r once $k \geq 2 = \lceil 5/2 \rceil - 1$, as predicted by (19).

Table 1: Observed convergence rates (exponent of $1/G$). Predicted values in parentheses.

Degree k	PSF profile (29)		$ x ^{5/2}$	
	value	derivative	value	derivative
1	2.01 (2)	1.00 (1)	1.93 (2)	0.97 (1)
2	3.10 (3)	2.00 (2)	2.42 (2.5)	1.46 (1.5)
3	4.18 (4)	3.09 (3)	2.42 (2.5)	1.45 (1.5)
5	6.49 (6)	5.34 (5)	2.42 (2.5)	1.45 (1.5)

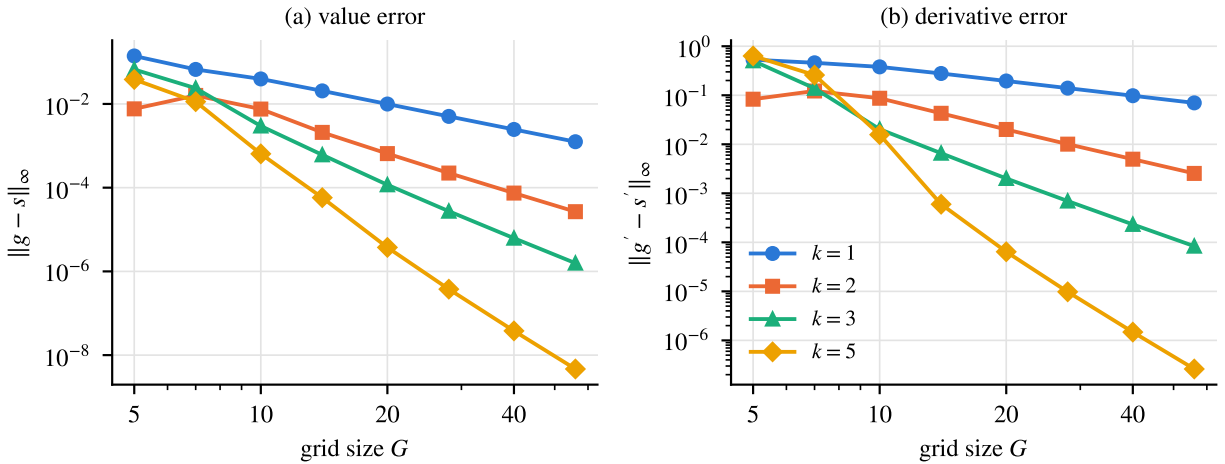


Figure 1: Approximation of the pixel-integrated PSF profile (29) by B-splines of degree k . (a) Sup-norm value error. (b) Sup-norm derivative error.

Grid allocation. Two targets of shape $[4, 1, 1]$ are used, $f(x) = \Phi(\sum_{i=1}^4 \phi(x_i))$ on $[-1, 1]^4$ with cubic splines ($p = 4$), $w_0 = 4$, $w_1 = 1$ and $z \in [-2.05, 2.05]$ at the second layer. In case D the inner functions

$\phi(x) = \frac{1}{2} \sin(3\pi x)$ are rough and $\Phi = \exp$ amplifies errors; in case E the inner functions $\phi(x) = \frac{1}{2} \sin(\pi x/2)$ are smooth and $\Phi(z) = \sin(3z)$ is the rough component. Rule (28) predicts $G_0^*/G_1^* = 2.95$ for case D and 0.36 for case E. Table 2 compares the allocation from (22) with an exhaustive search over all integer splits of the budget and with the uniform grid. In all four settings the rule lies within one grid interval of the exhaustive optimum, and it reduces the error by a factor of 1.5–1.8 (case D) and 11.8 (case E) relative to the uniform grid.

Table 2: Grid allocation at fixed coefficient budget $B = 4G_0 + G_1$. Errors are $\|f - \hat{f}\|_\infty$.

Case	B	Rule (22)			Exhaustive search		Uniform	
		G_0	G_1	error	G_0	error	error	E_u/E
D	120	28	8	4.07×10^{-3}	28	4.07×10^{-3}	6.11×10^{-3}	1.50
D	240	55	20	2.07×10^{-4}	56	2.01×10^{-4}	3.78×10^{-4}	1.83
E	120	18	48	8.67×10^{-6}	18	8.67×10^{-6}	1.02×10^{-4}	11.8
E	240	36	96	5.14×10^{-7}	35	5.01×10^{-7}	6.08×10^{-6}	11.8

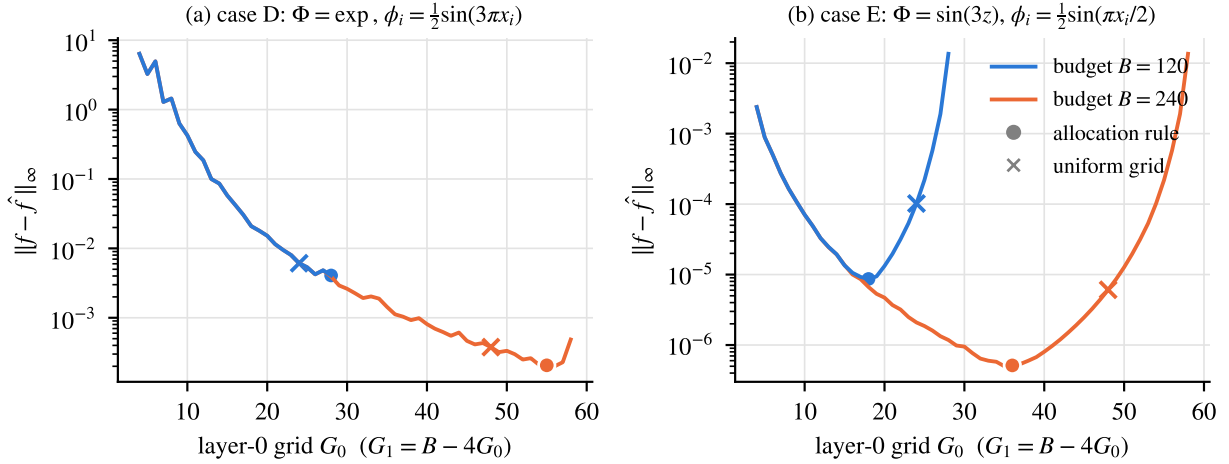


Figure 2: Error as a function of the split of a fixed budget between the two layers. Circles mark the allocation (22); crosses mark the uniform grid. In case D the two budget curves overlap where they share a range of G_0 , because the first layer dominates the error.

Weakest-link degree. For case E with equal grids $G_0 = G_1 = G$, the error of $\partial \hat{f} / \partial x_1$ was measured for five degree pairs (Figure 3). The observed exponents are 3.03, 2.20, 1.00, 1.00 and 1.00 for $(k_0, k_1) = (3, 3), (2, 3), (1, 3), (3, 1)$ and $(1, 1)$, against the predicted $\min(k_0, k_1) = 3, 2, 1, 1, 1$ of (17). Replacing either layer by a linear spline reduces the rate to first order, and the position of the linear layer does not matter.

7 Application to the KAN-SMLM Localizer

The results give three design rules for the localizer, to be tested on the existing models in the next phase of the work. First, by (20) and Corollary 1, every layer should use $k_l \geq 2$, with cubic splines as the default; a piecewise-linear layer anywhere limits the attainable gap to the Cramér–Rao bound. Second, inter-layer normalization fixes R_l and keeps Assumption 1 valid, which is required for both theorems. Third, grids

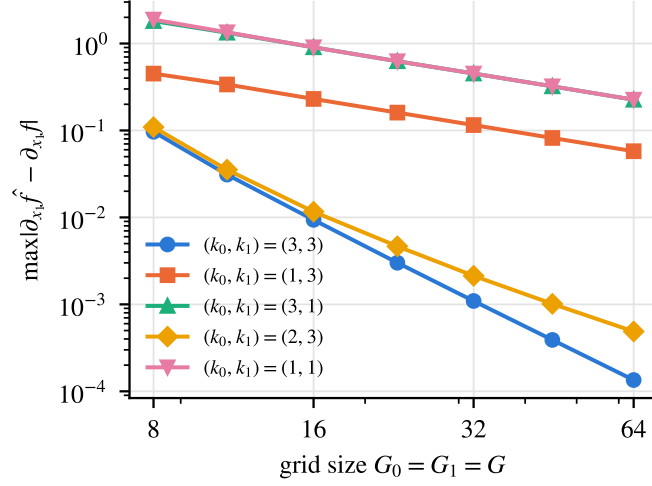


Figure 3: Sup-norm error of $\partial \hat{f} / \partial x_1$ for case E and mixed degrees (k_0, k_1) . The curves for $(3, 1)$ and $(1, 1)$ coincide.

should be allocated by (22) using Λ_l and $M_l^{(p)}$ estimated from a pilot model, instead of a common G for all layers.

The analysis bounds approximation error only. It does not address whether gradient training reaches the best spline coefficients, which recent comparisons identify as a frequent limitation of KANs [8, 9]. Theorem 3 is stated for a common degree; the same Karush–Kuhn–Tucker argument applies to mixed degrees, but the solution then has no closed form. Approximation rates for KANs in Sobolev and Besov norms beyond the uniform-norm setting used here are given in [6, 7].

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